NINGYUAN YANG
YOURAN WANG
ANDRANIK PAMBUKHCHYAN
BINGHAN CHENG

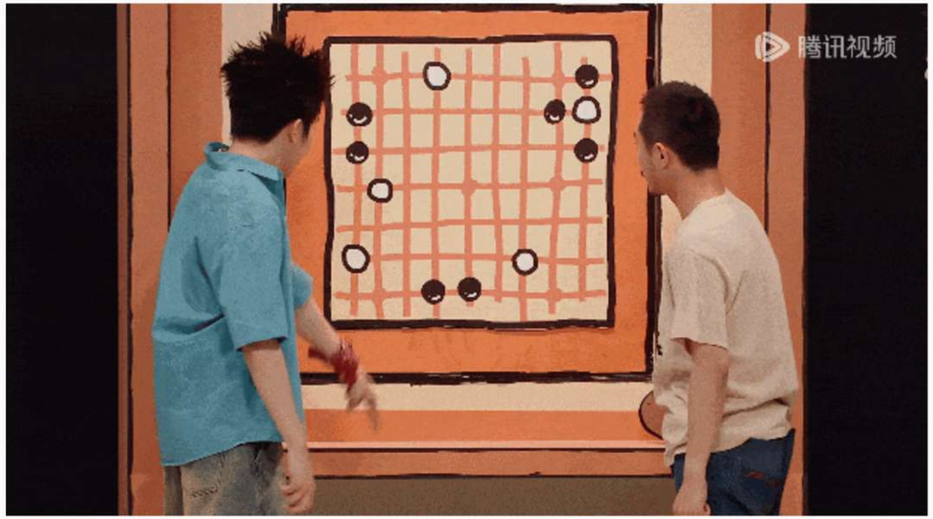
Skill Gobang

CS307 BK PROJECT

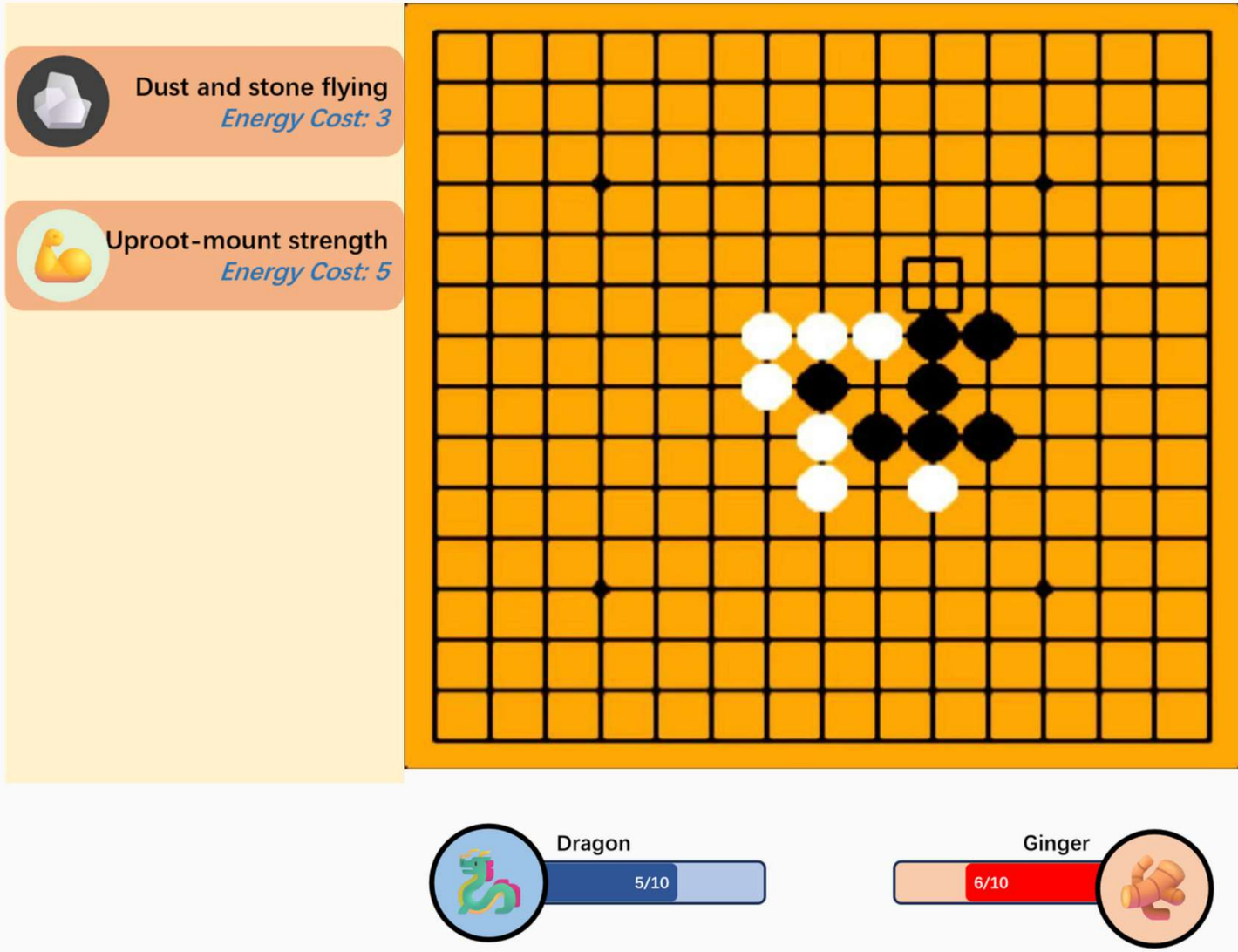
Outline

- Quick Review
- Division of Labor
- Update & Plan
- Timeline

Quick Review



技能五子棋*(Skill Gobang), 2025, Zhang et al.*

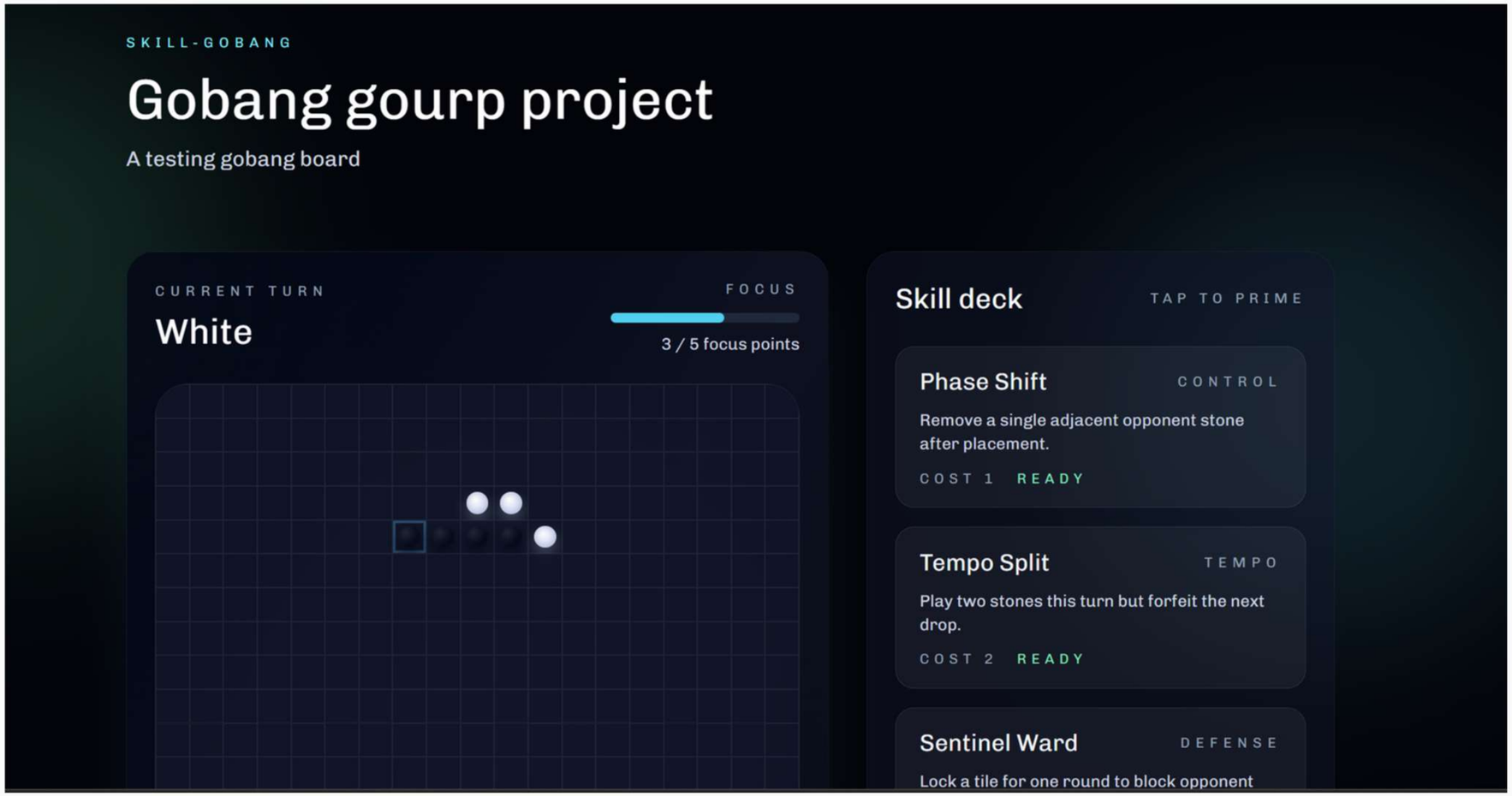


Division of Labor

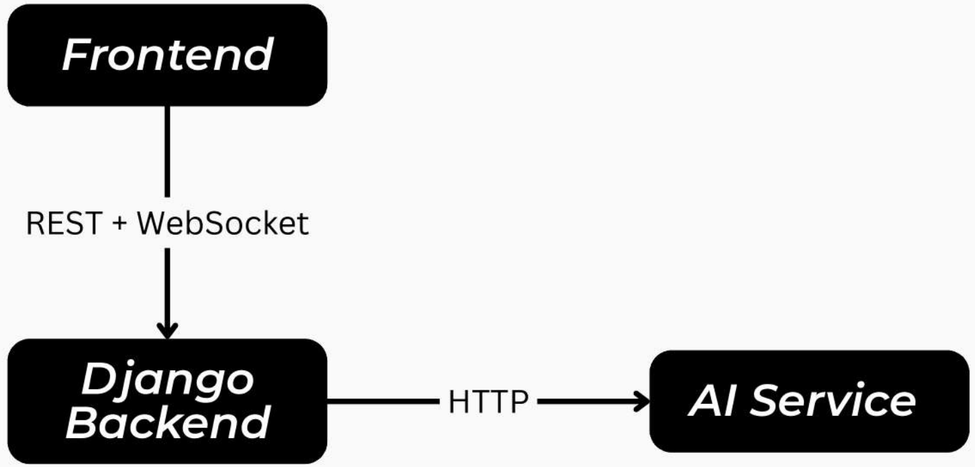
Frontend	Youran Wang
Backend	Binghan Cheng
AI opponent Design	Ningyuan Yang
Test	Andranik Pambukhchyan

Frontend design

The front end is built with a Next.js architecture and supports user interaction functionalities.



Backend Design



- *Auth, game state, skills*
- *Django Channels (WS)*
- *DRF (REST)*

- /api/ai/recommend
- /api/ai/top-moves

Field	Type
display_name	CharField(50)
games_played	IntegerField
games_won	IntegerField
games_lost	IntegerField
games_drawn	IntegerField

User

GamePlayer

Skill

Move

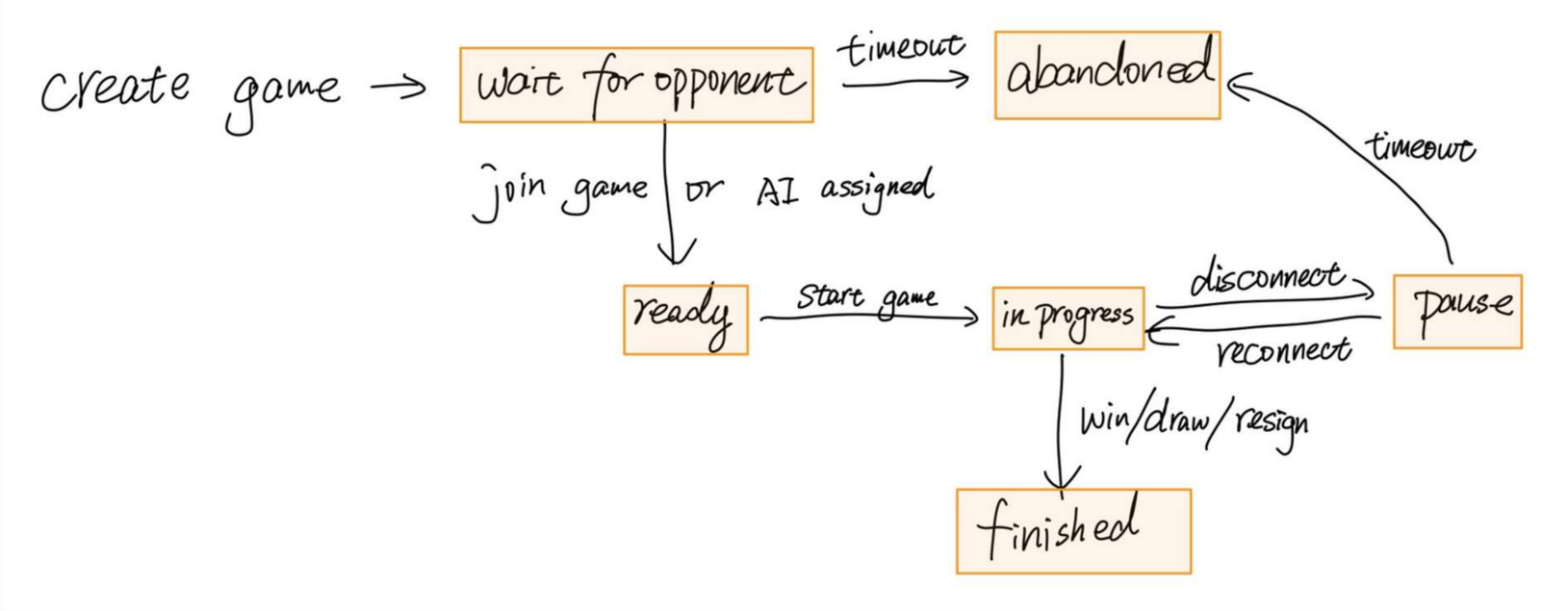
SkillUsage

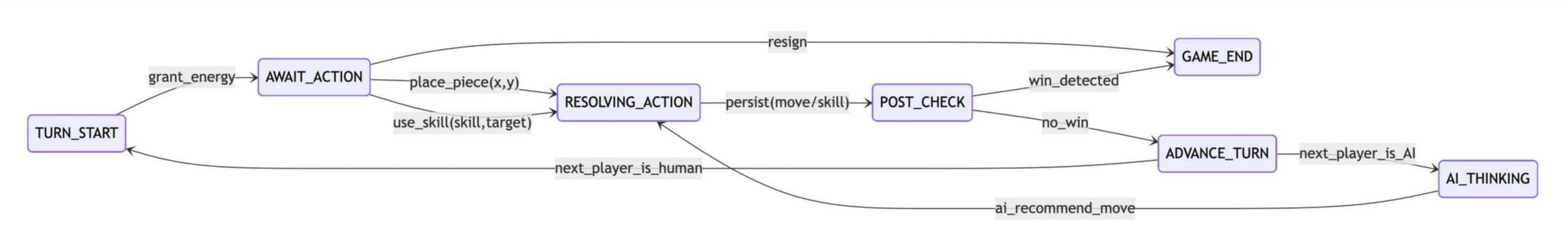
.....

Field	Type
id	UUIDField
mode	CharField
status	CharField
board_state	JSONField
blocked_positions	JSONField
current_turn	CharField
turn_number	IntegerField
extra_turns_remaining	IntegerField
winner	CharField, nullable
win_reason	CharField, nullable

Game

FSM Design for Game





AI opponent Design

Introducing LLM-based AI opponents in Skill-Gobang presents the following challenges:

- Relying solely on LLM: Severe hallucinations
- Integrating with Goban solver software: Unable to handle non-standard positions or execute skills.

AI opponent Design

Solution: MCP-Based Hybrid Approach Combining LLM & Gobang Solver

1. Write a script to provide solutions for Gobang games played under non-standard rules.
2. Wrap scripts into tools that can be invoked by LLMs via MCP
3. Each tern, the LLM may choose to use skills or call tools

输入参数：

```
{
  "myPieces": [{"x": 9, "y": 9}, {"x": 10, "y": 9}],
  "opponentPieces": [{"x": 9, "y": 10}],
  "blockedPositions": [{"x": 11, "y": 9}],
  "boardSize": 19
}
```

输出：

```
{
  "success": true,
  "recommendedMove": {"x": 8, "y": 9},
  "message": "推荐落子位置: (8, 9)"
}
```

Sample input & Output

Skill Design

Polarity reversal

- Energy Cost: 8
- Missing the next move, switch the colors of the black and white pieces on the board.

Still as a pond

- Energy Cost: 6
- Prevent the opponent from taking their next 2 turns.

Outline

- Phase 1: Plan & organize
  - Agree on the final feature list and what “done” means
  - Assign tasks clearly to each person
- Phase 2: Build the main structure
  - Set up the basic project so the core flow works end-to-end
- Phase 3: Finish key features
  - Implement the remaining features and make everything connect smoothly
- Phase 4: Test & fix
  - Test as a team, find bugs, and fix the important ones first
- Phase 5: Polish & prepare presentation
  - Improve the demo reliability, finalize slides, and rehearse the presentation

AI Usage & Human Contribution

Human Role:
Front-end and back-end logic design.
AI rule formulation and coding.
Test script development.
Skills Design

AI Role:
Programming bug fix
Wrap script to MCP Tools
Strategic move suggestions

Quiz

Q1. What is the main idea of “Skill Gobang”?

- A) A normal Gobang game with a bigger board
- B) Gobang where players can use special skills during the game
- C) A card game based on Gobang
- D) A multiplayer shooter game

Q2. Why is it difficult to use only an LLM (chatbot) as the AI opponent?

- A) It runs too fast
- B) It may hallucinate and make unreliable moves
- C) It cannot display graphics
- D) It uses too little memory

Q3. Why is it difficult to use only a traditional Gobang solver for the AI?

- A) It can’t play regular Gobang
- B) It can’t handle the added skills and non-standard situations well
- C) It cannot run on computers
- D) It requires internet access

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